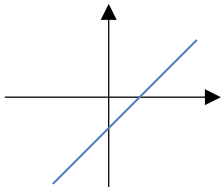
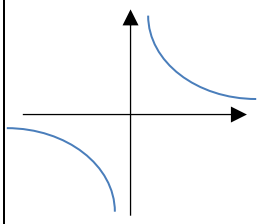
Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	12	**1**	
1(b)	8	**1**	
1(c)	5	**1**	
1(d)	$\sqrt{7}$	**1**	
2(a)	1 | 3 5 8 2 | 1 4 6 8 3 | 6 8 8 4 | 5 8	**2**	**M1** for correct but not ordered or for two correct rows ordered
2(b)	27	**1**	
3	$\begin{pmatrix} -4 \\ 3 \end{pmatrix}$	**1**	
4	165	**2**	**M1** for $\frac{(24-2)\times 180}{24}$ or $180-\frac{360}{24}$
5	$\frac{15}{28}\times\frac{7}{4}$ or $\frac{15}{28}\div\frac{16}{28}$ oe	**M1**	
	$\frac{15}{16}$ cao	**A2**	**A1** for $\frac{105}{112}$ oe
6	7.2	**3**	**M1** for $5\times 8+6\times 5+7\times 11+8\times 7+9\times 5+10\times 4$ **M1dep** for ÷ 40
7(a)	45.9	**2**	**M1** for $0.5\times 8.5\times 10.8$ oe
7(b)	33[.0] or 33.04…	**3**	**M2** for $8.5+10.8+\sqrt{8.5^2+10.8^2}$ oe or **M1** for $8.5^2+10.8^2$ oe
8	2.98×10^{-3}	**1**	
9(a)	$3x(x-4y)$ final answer	**2**	**B1** for $3(x^2-4xy)$ or $x(3x-12y)$

Question	Answer	Marks	Partial Marks
9(b)	$m^2 - m - 6$ final answer	**2**	**M1** for 3 terms from $m^2, -3m, +2m, -6$
10(a)	Correct sketch 	**1**	Line with positive gradient and negative y intercept
10(b)	Correct sketch 	**2**	**B1** for only one branch or attempt at correct shape
11(a)	Rotation 90° clockwise oe (0, 2)	**3**	**B1** for each
11(b)	Reflection $y = x$	**2**	**B1** for each
11(c)	Enlargement [sf] $\frac{1}{2}$ (4, 6)	**3**	**B1** for each
12	229 500 cao	**3**	**B2** for 229 460… OR **M1** for $250\,000 \times \left(1 - \frac{1.7}{100}\right)^5$ oe **B1** for *their* more accurate answer correctly rounded to the nearest 100
13	$2.\dot{6} - 0.2\dot{6}$ oe	**M1**	
	$\frac{4}{15}$ oe fraction nfww	**A1**	If **M0** scored **SC1** for $\frac{k}{90}$
14(a)	11.5	**1**	
14(b)(i)	12	**1**	
14(b)(ii)	8.5	**1**	